ASSESSMENT TOOL 1-CASE STUDY

Course Code: DSA0504	**Course Title: Query Processing for Data Science**
CO Assessed: CO4	**Assessment: Tool 1 – Case Study**
Weightage: 35	**Total Marks: 35 Marks**
CO4 Statement: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.	
Student Name:J.Subhaashree	**Reg. No.:192424008**
Section / Batch:2024	**Date of Submission:10/8/26**
Faculty In-charge: Dr. R.Mahesh Muthulakshmi	**Project Mode: Individual Submission**

CO4 – AT1 CASE STUDY ANALYSIS

Case Study 2 - E-Commerce Customer Database

Introduction

An online retail company maintains a customer database containing information about more than 50,000 customers. The database is used for marketing campaigns, customer segmentation, and sales analysis.

However, poor-quality data can reduce the effectiveness of marketing campaigns and produce inaccurate customer analysis. The identified problems include duplicate customer records, different phone number formats, incomplete addresses, typing errors in email addresses, and inconsistent capitalization of customer names.

Data wrangling is therefore required to clean, standardize, validate, and organize the customer information before it is used for analysis.

Problem Statement

The marketing department wants to improve campaign effectiveness by ensuring that customer information is accurate and consistent.

The major data quality problems are:

- Duplicate customer records
- Different phone number formats
- Incomplete addresses
- Incorrect or mistyped email addresses
- Inconsistent capitalization of customer names

Objectives

- To identify data quality problems in an e-commerce customer database.
- To classify the identified problems.
- To clean and standardize customer data using Python.
- To demonstrate duplicate removal and missing-value handling.
- To standardize names, phone numbers, and email addresses.
- To understand how clean data improves customer analytics.
- To propose a workflow for maintaining data quality.

Task 1 – Effect of Poor-Quality Customer Data on Marketing Decisions

Aim

To analyze how poor-quality customer data can affect marketing decisions.

Code

```
import pandas as pd

# Sample customer data
data = {
    "Customer_ID": [101, 102, 103, 104, 105, 101],
    "Name": [
        "Rahul Kumar",
        "Priya Sharma",
        "Arun Kumar",
        "Meena Rao",
        "John Peter",
        "Rahul Kumar"
    ],
    "Email": [
        "rahul@gmail.com",
        "priya@gmail.com",
        "arun@gmail.com",
        "meena@gmail.com",
        "john@gmail.con",
        "rahul@gmail.com"
    ],
    "Address": [
        "Chennai",
        "Bangalore",
        None,
        "Chennai",
        "Mumbai",
        "Chennai"
    ]
}

df = pd.DataFrame(data)

print("CUSTOMER DATA")
print(df)

# Analyze data quality
```

```
print("\n--- DATA QUALITY ANALYSIS ---")

print("\nDuplicate Records:")
print(df[df.duplicated("Customer_ID", keep=False)])

print("\nMissing Values:")
print(df.isnull().sum())

print("\nInvalid/Incorrect Emails:")
print(df[~df["Email"].str.contains(r"^[\w\.-]+@[\w\.-]+\.\w+$",
regex=True)])

print("\nTotal Customers:", len(df))
print("Unique Customers:", df["Customer_ID"].nunique())
```

Output

```
CUSTOMER DATA
   Customer_ID          Name            Email    Address
0          101   Rahul Kumar  rahul@gmail.com    Chennai
1          102  Priya Sharma  priya@gmail.com  Bangalore
2          103    Arun Kumar   arun@gmail.com       None
3          104     Meena Rao  meena@gmail.com    Chennai
4          105    John Peter   john@gmail.con     Mumbai
5          101   Rahul Kumar  rahul@gmail.com    Chennai

--- DATA QUALITY ANALYSIS ---

Duplicate Records:
   Customer_ID         Name            Email  Address
0          101  Rahul Kumar  rahul@gmail.com  Chennai
5          101  Rahul Kumar  rahul@gmail.com  Chennai

Missing Values:
Customer_ID    0
Name           0
Email          0
Address        1
dtype: int64

Invalid/Incorrect Emails:
Empty DataFrame
Columns: [Customer_ID, Name, Email, Address]
Index: []

Total Customers: 6
Unique Customers: 5
```

Case Study Analysis

The e-commerce company maintains customer information for more than 50,000 customers. This information is used for marketing campaigns, customer segmentation, and sales analysis. However, the presence of duplicate records, incomplete addresses, inconsistent phone numbers, and incorrect email addresses can reduce the quality of marketing decisions.

Duplicate records may cause the same customer to be counted multiple times and may result in repeated promotional messages. Incorrect email addresses can prevent marketing messages from reaching customers. Incomplete addresses can affect location-based marketing, while inconsistent customer names can make it difficult to identify and group customers correctly.

Therefore, poor-quality customer data can lead to **ineffective marketing campaigns, incorrect customer segmentation, wasted resources, and inaccurate business analysis**.

Key Analysis

- Duplicate records → Incorrect customer count
- Incorrect emails → Failed communication
- Incomplete addresses → Poor location-based marketing
- Inconsistent names → Incorrect customer identification
- Inconsistent phone numbers → Communication difficulties

Task 2 – Identification and Classification of Data Quality Issues

Aim

To identify and classify different data quality issues in an e-commerce customer database.

```
import pandas as pd

data = {
    "Customer_ID": [101, 102, 103, 104, 105, 101],
    "Name": [
        "RAHUL KUMAR",
        "priya sharma",
        "ARUN KUMAR",
        "Meena Rao",
        "JOHN PETER",
        "RAHUL KUMAR"
    ],
    "Phone": [
        "9876543210",
        "+91-98765-43211",
        "98765 43212",
        "+91 9876543213",
        "9876543214",
        "9876543210"
    ],
    "Email": [
        "rahul@gmail.com",
        "PRIYA@GMAIL.COM",
        "arun@gmail.com",
        "meena@gmail.com",
        "john@gmail.con",
        "rahul@gmail.com"
    ],
    "Address": [
        "Chennai",
        "Bangalore",
        None,
        "Chennai",
        "Mumbai",
        "Chennai"
    ]
```

```
}

df = pd.DataFrame(data)

print("DATA QUALITY ISSUES")
print("---------------------")

# Duplicate records
print("\n1. Duplicate Records")
print(df[df.duplicated("Customer_ID", keep=False)])

# Missing values
print("\n2. Missing Values")
print(df[df.isnull().any(axis=1)])

# Phone format inconsistency
print("\n3. Phone Number Formats")
print(df["Phone"])

# Name capitalization inconsistency
print("\n4. Name Capitalization")
print(df["Name"])

# Email capitalization
print("\n5. Email Format")
print(df["Email"])
```

OUTPUT

```
DATA QUALITY ISSUES
--------------------

1. Duplicate Records
   Customer_ID         Name       Phone           Email  Address
0          101  RAHUL KUMAR  9876543210  rahul@gmail.com  Chennai
5          101  RAHUL KUMAR  9876543210  rahul@gmail.com  Chennai

2. Missing Values
   Customer_ID        Name       Phone          Email Address
2          103  ARUN KUMAR  98765 43212  arun@gmail.com    None

3. Phone Number Formats
0         9876543210
1    +91-98765-43211
2        98765 43212
3     +91 9876543213
4         9876543214
5         9876543210
Name: Phone, dtype: object

4. Name Capitalization
0     RAHUL KUMAR
1    priya sharma
2      ARUN KUMAR
3       Meena Rao
4      JOHN PETER
5     RAHUL KUMAR
Name: Name, dtype: object

5. Email Format
0    rahul@gmail.com
1    PRIYA@GMAIL.COM
2     arun@gmail.com
3    meena@gmail.com
4     john@gmail.con
5    rahul@gmail.com
Name: Email, dtype: object
```

Case Study Analysis

The case study identifies several data quality problems in the customer database. These problems can be classified into duplicates, missing values, format inconsistencies, and accuracy issues.

The company has duplicate customer records because some customers register multiple times. Phone numbers are stored in different formats, addresses are incomplete, email addresses contain typing errors, and customer names have inconsistent capitalization.

Classification

Issue	Category	Effect
Duplicate customer records	Duplicate	Incorrect customer count
Different phone formats	Format inconsistency	Difficult communication
Incomplete addresses	Missing value	Incomplete customer profiles
Email typing errors	Accuracy issue	Failed email campaigns
Different name capitalization	Format inconsistency	Difficult customer matching

Analysis

Classifying the problems helps the company select the appropriate cleaning technique for each issue. For example, duplicate records can be removed, missing values can be handled, and inconsistent formats can be standardized.

Task 3 – Recommended Data Cleanup Techniques

Aim

To clean and standardize customer data using Python and Pandas.

```
import pandas as pd

import re


data = {

    "Customer_ID": [101, 102, 103, 104, 105, 101],

    "Name": [

        "RAHUL KUMAR",

        "priya sharma",

        "ARUN KUMAR",

        "Meena Rao",

        "JOHN PETER",

        "RAHUL KUMAR"

    ],

    "Phone": [

        "9876543210",

        "+91-98765-43211",

        "98765 43212",

        "+91 9876543213",

        "9876543214",

        "9876543210"

    ],
```

```
    "Email": [
        "rahul@gmail.com",
        "PRIYA@GMAIL.COM",
        "arun@gmail.com",
        "meena@gmail.com",
        "john@gmail.con",
        "rahul@gmail.com"
    ],
    "Address": [
        "Chennai",
        "Bangalore",
        None,
        "Chennai",
        "Mumbai",
        "Chennai"
    ]
}

df = pd.DataFrame(data)

print("ORIGINAL DATA")
print(df)

# 1. Standardize names
df["Name"] = df["Name"].str.strip().str.title()
```

```python
# 2. Clean phone numbers
df["Phone"] = df["Phone"].str.replace(r"\D", "", regex=True)

# Remove country code 91
df["Phone"] = df["Phone"].str.replace(r"^91", "", regex=True)

# 3. Standardize emails
df["Email"] = df["Email"].str.strip().str.lower()

# 4. Handle missing addresses
df["Address"] = df["Address"].fillna("Unknown")

# 5. Remove duplicate customer IDs
df = df.drop_duplicates(subset="Customer_ID", keep="first")

print("\nCLEANED DATA")
print(df)
```

OUTPUT

```
ORIGINAL DATA
   Customer_ID          Name           Phone            Email    Address
0          101    RAHUL KUMAR       9876543210  rahul@gmail.com    Chennai
1          102  priya sharma  +91-98765-43211  PRIYA@GMAIL.COM  Bangalore
2          103    ARUN KUMAR      98765 43212   arun@gmail.com       None
3          104     Meena Rao   +91 9876543213  meena@gmail.com    Chennai
4          105    JOHN PETER       9876543214   john@gmail.con     Mumbai
5          101    RAHUL KUMAR       9876543210  rahul@gmail.com    Chennai

CLEANED DATA
   Customer_ID          Name       Phone            Email    Address
0          101    Rahul Kumar  9876543210  rahul@gmail.com    Chennai
1          102  Priya Sharma  9876543211  priya@gmail.com  Bangalore
2          103    Arun Kumar  9876543212   arun@gmail.com    Unknown
3          104     Meena Rao  9876543213  meena@gmail.com    Chennai
4          105    John Peter  9876543214   john@gmail.con     Mumbai
```

Case Study Analysis

Data cleanup is required to transform the raw customer database into a consistent and reliable dataset. The first step is to identify duplicate records and remove unnecessary copies. Missing information such as addresses should be identified and handled appropriately.

Customer names should be converted into a consistent format, such as title case. Phone numbers should be converted into a standard numerical format by removing spaces, hyphens, and unnecessary country-code formatting. Email addresses should be converted to lowercase and validated.

Recommended Techniques

1. Duplicate Removal

Identify customers using Customer ID, email, phone number, or a combination of fields and remove duplicate records.

2. Missing Value Handling

Identify missing information and fill it using verified information where possible. Otherwise, values such as `Unknown` can be used.

3. Name Standardization

Convert names such as:

`RAHUL KUMAR` → `Rahul Kumar`

4. Phone Standardization

Convert different formats into one standard format:

`+91-98765-43210` → `9876543210`

5. Email Standardization

Convert:

`RAHUL@GMAIL.COM` → `rahul@gmail.com`

These techniques make the customer database more consistent and suitable for analysis.

Task 4 – How Standardization Improves Customer Analytics

Aim

To demonstrate how standardization improves the consistency and usability of customer data.

```
import pandas as pd

data = {
    "Customer_ID": [101, 102, 103, 104],
    "Name": [
        "RAHUL KUMAR",
        "rahul kumar",
        "Rahul Kumar",
        "PRIYA SHARMA"
    ],
    "Email": [
        "RAHUL@GMAIL.COM",
        "rahul@gmail.com",
        "Rahul@gmail.com",
        "PRIYA@GMAIL.COM"
    ]
}

df = pd.DataFrame(data)

print("BEFORE STANDARDIZATION")
print(df)

# Standardize names
df["Name"] = df["Name"].str.strip().str.title()

# Standardize emails
df["Email"] = df["Email"].str.strip().str.lower()

print("\nAFTER STANDARDIZATION")
print(df)

# Count unique names before/after
print("\nNumber of unique standardized names:")
print(df["Name"].nunique())
```

```
print("\nStandardized customer names:")
print(df["Name"].unique())
```

OUTPUT

```
BEFORE STANDARDIZATION
   Customer_ID          Name              Email
0          101   RAHUL KUMAR  RAHUL@GMAIL.COM
1          102   rahul kumar  rahul@gmail.com
2          103   Rahul Kumar  Rahul@gmail.com
3          104  PRIYA SHARMA  PRIYA@GMAIL.COM

AFTER STANDARDIZATION
   Customer_ID          Name              Email
0          101   Rahul Kumar  rahul@gmail.com
1          102   Rahul Kumar  rahul@gmail.com
2          103   Rahul Kumar  rahul@gmail.com
3          104  Priya Sharma  priya@gmail.com

Number of unique standardized names:
2

Standardized customer names:
['Rahul Kumar' 'Priya Sharma']
```

Case Study Analysis

Standardization ensures that the same type of information is represented in a consistent format throughout the database. Without standardization, the same customer may appear as different records because of differences in capitalization, phone-number formatting, or email formatting.

For example:

RAHUL KUMAR

Rahul Kumar

rahul kumar

These values can be standardized as:

Rahul Kumar

Similarly:

RAHUL@GMAIL.COM

rahul@gmail.com

Rahul@gmail.com

can be standardized as:

rahul@gmail.com

Benefits to Customer Analytics

Standardization helps in:

- Accurate customer identification
- Customer segmentation
- Duplicate detection
- Customer behavior analysis
- Marketing campaign targeting
- Sales analysis
- Generation of reliable reports

Therefore, standardized data provides a more accurate representation of customers and improves the reliability of analytical results.

Task 5 – Workflow for Maintaining Data Quality

Code

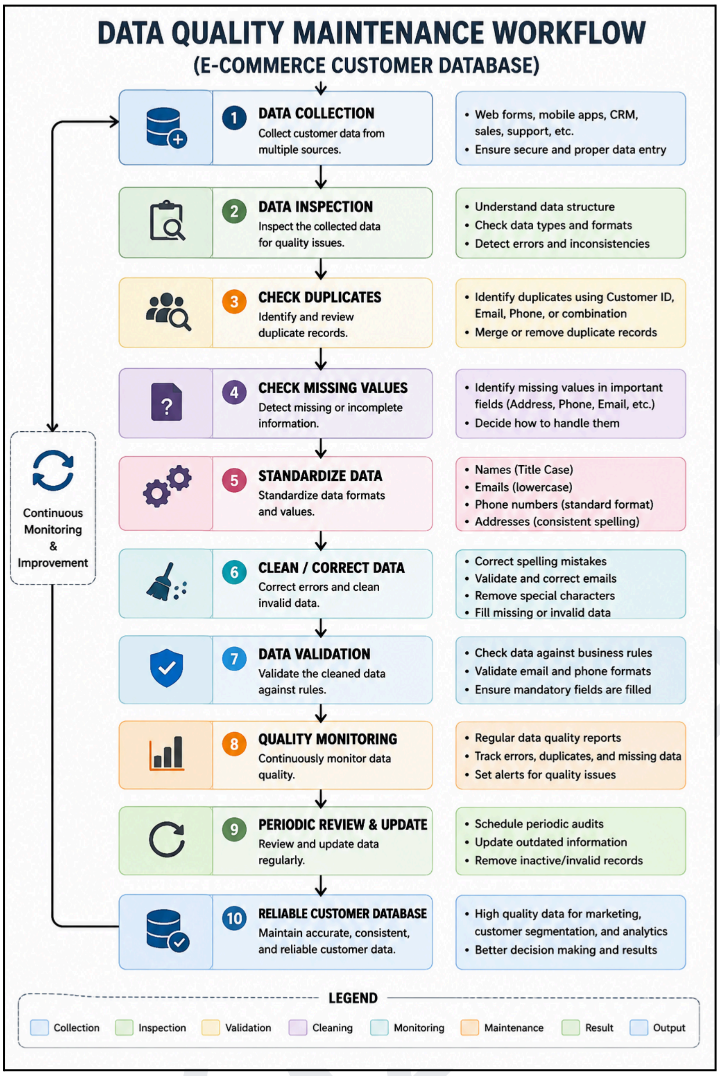
```
import pandas as pd

data = {
    "Customer_ID": [101, 102, 103, 101],
    "Name": ["RAHUL KUMAR", "Priya Sharma", "ARUN KUMAR", "RAHUL
KUMAR"],
    "Email": [
        "RAHUL@GMAIL.COM",
        "priya@gmail.com",
        "arun@gmail.com",
        "RAHUL@GMAIL.COM"
    ],
    "Address": ["Chennai", "Bangalore", None, "Chennai"]
}

df = pd.DataFrame(data)

print("STEP 1: DATA COLLECTION")
print(df)

# Step 2: Detect duplicates
print("\nSTEP 2: CHECK DUPLICATES")
print("Duplicate records:", df.duplicated("Customer_ID").sum())

# Step 3: Check missing values
print("\nSTEP 3: CHECK MISSING VALUES")
print(df.isnull().sum())

# Step 4: Standardize data
df["Name"] = df["Name"].str.strip().str.title()
df["Email"] = df["Email"].str.strip().str.lower()

# Step 5: Handle missing values
df["Address"] = df["Address"].fillna("Unknown")

# Step 6: Remove duplicates
df = df.drop_duplicates(subset="Customer_ID")

# Step 7: Final validation
```

```
print("\nSTEP 4: CLEANED DATA")
print(df)

print("\nSTEP 5: FINAL VALIDATION")
print("Duplicate records:", df.duplicated("Customer_ID").sum())
print("Missing values:")
print(df.isnull().sum())
```

OUTPUT

```
STEP 1: DATA COLLECTION
   Customer_ID          Name            Email    Address
0          101   RAHUL KUMAR  RAHUL@GMAIL.COM    Chennai
1          102  Priya Sharma  priya@gmail.com  Bangalore
2          103    ARUN KUMAR   arun@gmail.com       None
3          101   RAHUL KUMAR  RAHUL@GMAIL.COM    Chennai

STEP 2: CHECK DUPLICATES
Duplicate records: 1

STEP 3: CHECK MISSING VALUES
Customer_ID    0
Name           0
Email          0
Address        1
dtype: int64

STEP 4: CLEANED DATA
   Customer_ID          Name            Email    Address
0          101   Rahul Kumar  rahul@gmail.com    Chennai
1          102  Priya Sharma  priya@gmail.com  Bangalore
2          103    Arun Kumar   arun@gmail.com    Unknown

STEP 5: FINAL VALIDATION
Duplicate records: 0
Missing values:
Customer_ID    0
Name           0
Email          0
Address        0
dtype: int64
```

Wokflow



Case Study Analysis

Maintaining data quality should be a continuous process rather than a one-time cleaning activity. The company should establish a systematic workflow beginning with data collection and inspection.

The collected data should first be checked for duplicates, missing values, incorrect formats, and inconsistent information. After identifying the problems, the data should be cleaned and standardized. The cleaned data should then be validated against predefined quality rules.

The database should also be monitored regularly so that newly introduced errors can be detected and corrected. This ensures that the customer database remains accurate and useful for future marketing and analytics activities.

Analysis

This workflow helps the company maintain high-quality customer information throughout the data lifecycle. Continuous monitoring is important because new customers, updates, and transactions can introduce new errors into the database.

Expected Benefits

- Improved data accuracy
- Reduced duplicate records
- Consistent customer information
- Better marketing campaigns
- Reliable customer segmentation
- More accurate business reports

Tools and Techniques

Tool/Technique	Purpose
Python	Data processing and automation
Pandas	Data cleaning and manipulation
Regular Expressions	Phone/email pattern processing
Data Validation	Checking data correctness
Duplicate Detection	Identifying repeated customers
Standardization	Creating consistent formats

Python and Pandas are particularly suitable because they can automate repetitive data-cleaning operations on large customer datasets.

Advantages of Maintaining Data Quality

Maintaining high-quality customer data provides several benefits:

1. Better marketing campaigns – campaigns can target the correct customers.
2. Reduced duplicate communication – customers are less likely to receive repeated messages.
3. Improved customer segmentation – customers can be grouped more accurately.
4. Better sales analysis – clean data produces more reliable reports.
5. Reduced operational cost – less time is spent correcting data manually.
6. Improved decision-making – management can make decisions using reliable information.
7. Better customer experience – accurate contact information improves communication.

Conclusion

The e-commerce customer database contains several data quality problems, including duplicate records, inconsistent phone number formats, incomplete addresses, email errors, and inconsistent capitalization. These problems can negatively affect marketing campaigns, customer segmentation, and sales analysis.

A systematic data-wrangling process can solve these issues by identifying, cleaning, standardizing, and validating the data. Python and Pandas can automate many of these operations efficiently.

Therefore, maintaining clean and standardized customer data is essential for accurate customer analytics, effective marketing campaigns, and better business decision-making.